

Let's Measure in practical situations: Grade 7

A. Title of card: Measurement of lengths of objects and conversion of units of lengths
B. Competency, Element and Performance Criteria: C 2.3: Use measurement in real life situations Element 2: Use measurement in real life Performance Criteria: Identify units of measurement, measure lengths of objects and convert units of lengths (metres to centimetres and vice versa)
C. Purpose of activity: This activity will develop students' ability to identify the attributes to be measured and carry out measurements of various objects in the environment.
D. Learning Outcomes: By the end of the lesson, students should be able to • Identify the attributes that need to be measured • Identify the appropriate unit of measurement • Measure lengths of objects using a measuring tool • Convert units of length (m to cm and vice versa)
E. Resources and materials: • Ruler, measuring tape 

F. Teaching trajectories/Implementation guidelines:

Bring a ruler and a measuring tape to measure different objects.

Put students in groups of 2-4

Steps:

1. Ask students to observe the different objects/items in the class (e.g., pen, rubber, coloured pencils, ruler, chair, table, cupboard) and discuss them within the group.
2. The teacher asks the following questions:
 - (i) How are the objects different?
 - (ii) How are they the same?
3. The teacher writes students' responses on the board (terms like heavy, light, long, short and colours are noted for specific objects/items).
4. The teacher asks which of these attributes can be measured?
5. The class will focus on only **one** attribute namely: **length**.
6. The teacher asks the students to measure the length of each object.
7. The students select the appropriate measurement tool to measure the objects (e.g., pencil, table, whiteboard).
8. The teacher now asks the students to measure the length of the class.
9. The teacher assists the students while measuring the length of the class.
 - (i) Is a ruler an appropriate tool?
 - (ii) Which measuring tool will be more appropriate?
10. Discuss the units of measure of lengths such as centimetres (cm) and metres (m).
11. Deduce the relationship between cm and m

$$1\text{ m} = 100\text{ cm}$$

The teacher takes the students out of the class and asks them to measure the length and width of different items (e.g., to measure the length and width of the playground).

The following activities are meant to initiate and consolidate the above teaching trajectories.

Activity 1: Estimating and measuring length

Estimating and Measuring Length



You will work either pairwise or in groups. Follow the instructions from your teacher carefully.

You will need a measuring tape.

- Estimate the following:
 - the length around your waist
 - the length of your right hand
 - the length of your left leg
- Using a measuring tape, measure the following:
 - the length around your waist
 - the length of your right hand
 - the length of your left leg
- What do you notice about the measurements?
- Which is the longest of the three body parts? Give its actual measurement.
- Which is the smallest of the three body parts? Give its actual measurement.





AN INTERESTING FACT

The arm span is the distance stretched from one fingertip of the person's arm to the other. The arms should be parallel to the ground at the shoulder level. It is believed that on average this distance might approximately be the height of the person.





CHECK THIS LINK

Here's an interactive game on measurement:
<http://mrmussbaum.com/sol/>



The terms 'long', 'short' and 'tall' refer to the length of objects. 'Long' is used to refer to horizontal distance while 'tall' is the term used for vertical distance.

Example:
The queue at the bank is **long**.
The State Bank of Mauritius is a **tall** building in Port Louis.
Tanveer is **short** while Aisha is **tall**. (We do not say that Aisha is **long**.)

Activity 2

Measure the length of AB and BC



(a) $AB =$ _____ cm

$BC =$ _____ cm

(b) Join A to C and measure its length

$AC =$ _____ cm

(c) AC is longer than AB by _____ cm.

(d) AC is longer than BC by _____ cm.



ACTIVITY: DISTINGUISH AMONG DIFFERENT UNITS OF LENGTH



With the help of your teacher, discuss on the following:

1. What do we need to measure things around us?
2. What units of length are you familiar with?
3. For each of the cases below, state which unit of length you would use.
 - (a) Tessa wants to buy some cloth to make a dress for her daughter. What must she do?
 - (b) I want to put a fence around my house. What must I do?



EXERCISE



Match each of the following length with its most appropriate unit to be used.



Cloth used to make a dress

cm



Length of a pen

m

Activity: Convert metre to centimetre and vice versa

Materials needed: some measuring tapes

Procedure:

- Put the students in pairs and give each pair a measuring tape.
- Ask them to measure one metre.
- Beneath the 1 m, they will find another measurement in cm.
- Show them that $1 \text{ m} = 100 \text{ cm}$.
- Ask them to find the number of centimetres in
 - (a) $\frac{1}{2} \text{ m}$
 - (b) $\frac{1}{4} \text{ m}$
 - (c) $\frac{3}{4} \text{ m}$
- Ask them to find the number of metres in
 - (a) 50 cm
 - (b) 150 cm
- Ask them to deduce the number of centimetres in (a) 2 m, (b) 5 m, (c) $6\frac{1}{2} \text{ m}$
- Explain to students that to convert m to cm, we multiply by 100 but to convert cm to m, we divide by 100.

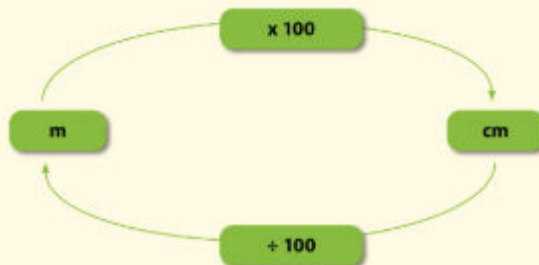
ACTIVITY: CONVERTING METRE INTO CENTIMETRE AND VICE-VERSA



With the help of your teacher and working in pairs, find the following:

- Measure 1 metre with the measuring tape.
- Find the corresponding measurement in centimetres.
- What do you notice?
- Find the number of centimetres in the following:
(a) $\frac{1}{2}$ m (b) $\frac{1}{4}$ m (c) $\frac{3}{4}$ m
- Find the number of metres in: (a) 50 cm (b) 150 cm.
- Find the number of centimetres in: (a) 2 m (b) 5 m (c) $6\frac{1}{2}$ m.
- What can you deduce?

To convert metre to centimetre and vice-versa, we do the following:





Convert into centimetres.

For example: $7\text{ m} = 7 \times 100\text{ cm}$
 $= 700\text{ cm}$

Convert into metres, giving your answer as a fraction where necessary.

For example: $550\text{ cm} = \frac{550}{100}\text{ m}$
 $= 5\frac{1}{2}\text{ m}$



EXERCISE



Try these.

1. Complete the table.

Metres	1	2	3	4	5	6
Centimetres	100	200				

2. Convert the following into centimetres.

(a) 4 m

(b) $6\frac{1}{2}\text{ m}$

(c) 2 m 5 cm

(d) $13\frac{3}{4}\text{ m}$

3. Convert the following into metres.

(a) 200 cm

(b) 450 cm

(c) 775 cm

(d) 125 cm

4. A table is 1.2 metres long and a bench is 95 cm long. What is their combined length in (i) centimetres and (ii) metres?

G. Assessment (Work out the above 4 questions to assess students' understanding of the conversion of units of lengths).

H. Extension of activity

Find the *Perimeter* of different shapes and items.